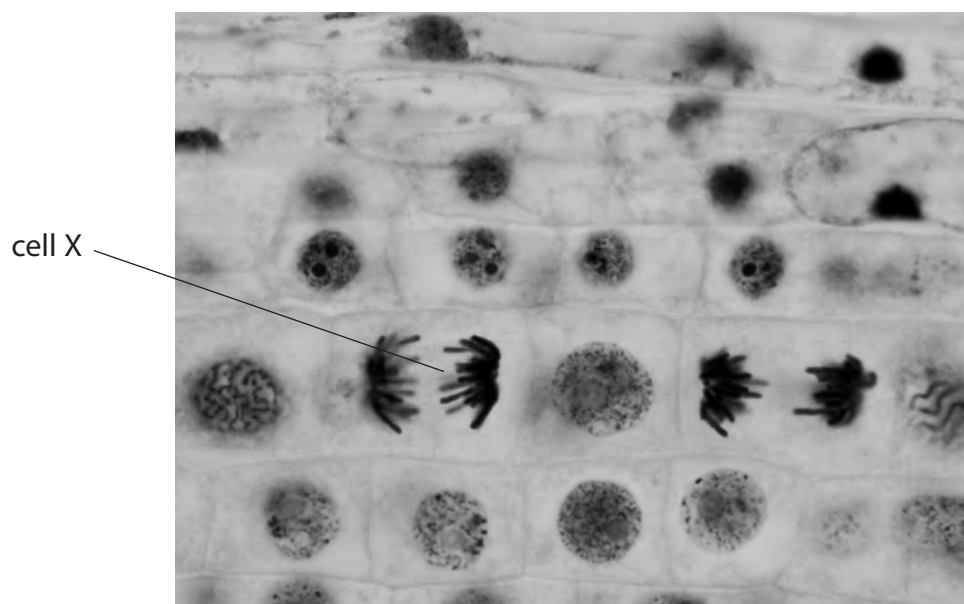


- 1 The photograph shows cells undergoing mitosis in the root of an onion plant.



(Source: © blickwinkel/Alamy Stock Photo)

- (a) Name the stage of mitosis shown in cell X.

(1)

(b) A student read a scientific report that stated:

'Cell division was reduced when plant cells were treated with a caffeine solution.'

(i) Describe an experiment to measure the mitotic index of cells from onion roots treated with different concentrations of caffeine solution.

(6)

- (ii) The scientific report suggested that caffeine affected the production of the cellulose molecules needed for cell division.

Describe the structure of a cellulose molecule.

(3)

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(Total for Question 1 = 10 marks)